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Diameter effect on critical heat flux

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ABSTRACT

The critical heat flux look-up table (CHF LUT) is widely used to predict CHF for various applications, including design and safety analysis of nuclear reactors. Using the CHF LUT for round tubes having inside diameters different from the reference 8 mm involves conversion of CHF to 8 mm. Different authors [Becker, K.M., 1965. An Analytical and Experimental Study of Burnout Conditions in Vertical Round Ducts, Aktiebolaget Atomenergie Report AE 177, Sweden; Boltenko, E.A., et al., 1989. Effect of tube diameter on CHF at various two phase flow regimes, Report IPE-1989; Biasi, L., Clerici, G.C., Garriba, S., Sala, R., Tozzi, A., 1967. Studies on Burnout, Part 3, Energia Nucleare, vol. 14, pp. 530–536; Groeneveld, D.C., Cheng, S.C., Doan, T., 1986. AECL-UO critical heat flux look-up table. Heat Transfer Eng., 7, 46–62; Groeneveld et al., 1996; Hall, D.D., Mudawar, I., 2000. Critical heat flux for water flow in tubes – II subcooled CHF correlations. Int. J. Heat Mass Transfer, 43, 2605–2640; Wong, W.C., 1996. Effect of tube diameter on critical heat flux, MaSC dissertation, Ottawa Carleton Institute for Mechanical and Aeronautical Engineering, University of Ottawa] have proposed several types of correlations or factors to describe the diameter effect on CHF. The present work describes the derivation of new diameter correction factor and compares it with several existing prediction methods.

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1. Introduction

The critical heat flux look-up table (CHF LUT) is widely used to predict CHF for various applications, including design and safety analysis of nuclear reactors. Using the CHF LUT for round tubes having inside diameters different from the reference 8 mm involves conversion of CHF to 8 mm. Different authors (Becker, 1965; Boltenko et al., 1989; Biasi et al., 1967; Groeneveld et al., 1986, 1992, 1996, 2005; Hall and Mudawar, 2000; Wong, 1996) have proposed several types of correlations or factors to describe the diameter effect on CHF. The present work describes the derivation of new diameter correction factor and compares it with several existing prediction methods.

2. Mechanistic description of the diameter effect

2.1. Subcooled CHF

For the same local conditions, the subcooled CHF generally increases with a reduction in tube inside diameter. This could be due to the reduced distance between the bubbly boundary layer and subcooled core for small diameter tubes: this reduced distance enhances the rate of vapour condensation within the boundary layer, thus increasing CHF. For the same pressure, mass flux and subcooling, the average flow velocity is similar, regardless the tube diameter, but smaller tubes result in larger radial velocity gradients. Higher velocity gradient enhances the rate of detachment of the growing bubbles, thus increasing the CHF. Furthermore, higher temperature gradients in smaller tubes enhance the heat transfer between bubbles and liquid, reducing bubble generation and increasing CHF.

2.2. Film dryout

At higher critical qualities, the observed experimental trends are explained through the annular flow model by considering the variation of entrainment, evaporation and deposition rates with diameter. In general the entrainment rate strongly depends on liquid film thickness, a thicker film produces higher amplitude surface waves thus enhancing the entrainment rate. For the same local quality, pressure and mass flux, smaller tubes have a thinner film and a smoother liquid film interface, thus reducing entrainment rate. Furthermore, a small diameter channel would have shorter heated length, hence there will be less time available for

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Table 1
Diameter effect and mechanisms, updated from Wong (1996).

	Phenomenon in small diameter tubes	Mechanisms for CHF increase
Subcooled and low quality flow	Smaller vapour bubble size	Better heat transfer because of core flow close to the wall
	Thinner vapour bubble boundary layer	
	Void fraction within bubble layer decreases	Re-entry of liquid replenishment easier
	Velocity gradient of two-phase boundary layer increases	Growing vapour bubble detaches easier before forming vapour patches at wall
	Temperature gradient of wall layer higher	Bubble generation slower due to high condensation rate at the tip of the bubble
	Interfacial friction inversely proportional with *Re* number	Increase friction at two phase boundary and increase heat transfer
High quality, annular flow	Flow of liquid layer at wall thinner	Boiling suppressed in the liquid film
	Lower entrainment rate	Higher liquid film thickness
	Temperature gradient of liquid film at wall increases	Reduced liquid superheat in the film, lower evaporation rate
	Rate of liquid deposition increases	Higher liquid film thickness
Zero flow/low flow	Smaller subcooled liquid volume	Lower bubble dissipation rate
	Higher liquid/vapour velocity gradients	Higher shear stress at interface liquid/vapour

entrainment. Table 1 summarizes the main phenomena and mechanisms for CHF enhancement in small diameter tubes.

2.3. Low flow or zero flow CHF

The CHF mechanism at zero and very low flows usually corresponds to the counter-current flow limitation (CCFL) or flooding, except for very large diameters when a pool type CHF occurs. For these flow conditions, experimental data suggest that higher CHF is obtained at higher diameters. Large diameters allow better development of the counter-current flow due to lower velocity gradients and shear stress, therefore the liquid flow limitation occurs at higher surface heat fluxes and thus CHF.

At zero/very low flows in larger channel diameters, the large volume of subcooled liquid will permit better bubble dissipation and condensation, hence it is less likely that large vapour slugs are formed that could otherwise bridge the heated channels and initiate CHF condition.

3. Prediction methods for diameter effect

To account for the diameter correction for the CHF LUT, Doroshchuk et al. (1975) and Groeneveld et al. (1986, 1992, 1996) have proposed the following equation:

$$\frac{CHF_D}{CHF_{D=8}} = \left(\frac{8}{D}\right)^n \tag{1}$$

Different values of the exponent n have been recommended: 1/2 (Doroshchuk et al., 1975; Groeneveld et al., 1996) and 1/3 (Groeneveld et al., 1986). These exponents are optimized from large databases and represent average values. Analysis of the available experimental data suggests that the exponent n can also be a function of pressure, mass flux, dryout quality, and diameter itself. For example, the exponent for the diameter term in the Biasi correlation is 0.4 for a tube of diameter less than 10 mm, and 0.6 otherwise.

For high pressure CHF data, Becker (1965) presented ratios of CHF for various values of tube diameter to CHF for a 10 mm tube. These ratios varied from 1.18 for a 4-mm tube and 0.90 for a 25-mm tube.

Based on the experimental database of IPPE, Obninsk and his own experiments using Freon, Boltenko et al. (1989) proposed a correlation, which takes into account the flow regimes and the CHF mechanisms. He used three different tubes of 8, 16 and 20 mm ID and observed that for the low quality region ($X_c < 0.2$) the CHF increases with tube ID, while high qualities showed a reverse trend. To quantitatively assess the diameter effect, he used the general correlation (1), where the exponent n varies according to the flow regimes, as presented in Table 2. Fortini and Veloso (2002) proposed a correlation for diameter correction, very similar to Boltenko.

Table 2 shows the most common diameter correction correlations from literature and their range of applicability.

4. Derivation of new prediction method for diameter effect on CHF

The basis for the assessment is to "separate" the effect of the diameter from the effect of other parameters. For this analysis the CHF is considered a local conditions function, as follows:

$$CHF = \Phi(P, G, X, D) \tag{2}$$

where Φ is an unknown real function.A number of 26,238 experimental data points, with diameter range from 3 to 45 mm has been considered.A slicing method (Durmayaz et al., 2004) was used to isolate the diameter effect: for each nominal LUT pressure $[(P_{i-1}+P_i)/2] < P < [(P_{i+1}+P_i)/2]$, nominal mass flux $[(G_{j-1}+G_j)/2] < G < [(G_{j+1}+G_j)/2]$ and nominal quality $[(X_{k-1}+X_k)/2] < X < [(X_{k+1}+X_k)/2]$, a CHF versus diameter plot was created. Hence, for each slice the unknown function $\Phi(P, G, X, D)$ turns into $\Psi(D)$, with P, G, X = constant. For such conditions, the study of Ψ with diameter is more straightforward.

The whole range of CHF LUT flow conditions was divided in 24 sections for pressure, 21 for mass flux and 23 for quality. For each slice, the experimental parameters are not exactly the same as the nominal values of the slices, because the experimental parameters fall in the slicing range. To be able to compare the values at the same flow conditions, a correction based on the slopes of the 2005 LUT was applied. Experimental values were modified based on their distances from the nominal value, with respect to all three directions, P, G and X, respectively:

$$q_c = q_{c_0} + \frac{\partial q_c}{\partial P}(P_0 - P_i) + \frac{\partial q_c}{\partial G}(G_0 - G_j) + \frac{\partial q_c}{\partial X}(X_0 - X_k) \tag{3}$$

where "0" index refers the experimental points and i, j, k refer the nominal values which characterize the slice with respect to P, G and X. The partial derivatives (slopes) were obtained from the 2005 LUT. The prediction errors are calculated for each data point in the database. All the errors are based on local conditions.

Although it was possible to produce 11592 slices ($21 \times 23 \times 24$), only those slices that contained CHF data at different diameters were considered, thus allowing the establishment of a trend. A sample of a slice at nominal P = 7000 kPa, G = 4000 kg m^{-2} s^{-1}, X_c = 0, is presented in Fig. 1. As shown in the figure, the reference diameter is 8 mm and the reference CHF at 8 mm is calculated by averaging all

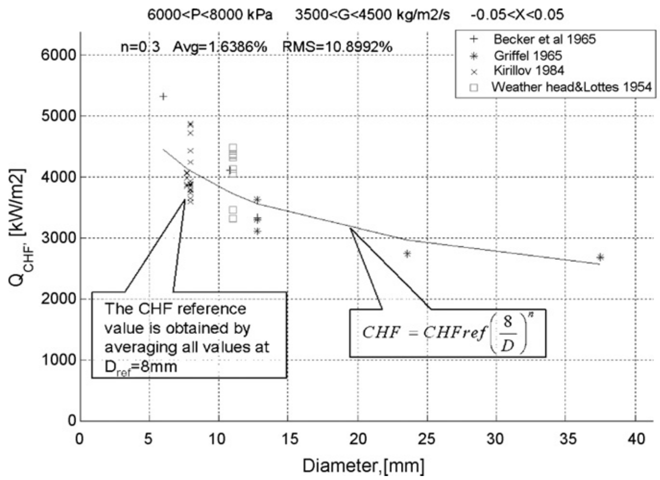

Fig. 1. Slice showing the typical trend of CHF versus diameter and the best fitting curve.

the CHF values at 8 mm. The prediction of CHF at other diameters was performed using Eq. (1).

The exponent n was varied until the RMS error calculated reached its minimum; for the above plot $n = 0.3$, produced a minimum RMS error of 10.0%. If the generated slice does not contain any data at $D = 8$ mm, then the reference diameter is assumed to be the diameter closest to 8 mm.

The general equation is:

$$\frac{CHF_{D1}}{CHF_{D2}} = \left(\frac{D_2}{D_1}\right)^n \quad \text{or} \quad CHF_{D1} = CHF_{D2}\, D_2^n \left(\frac{1}{D_1}\right)^n \tag{4}$$

To examine more closely whether n remained reasonably constant, the whole range of P, G and X was subdivided based on the observed trends and the following considerations:

- Because the pressure has the least impact on CHF and n, only two pressure ranges ($P < 14{,}000$ kPa and $P > 14{,}000$ kPa) have been considered; this approach is consistent with Boltenko's pressure subdivision;
- Experimental data and some papers (Mishima et al., 1985; Chang et al., 1991) revealed that at low flows (<250 kg m^{-2} s^{-1}) the CHF mechanisms are different compared to other flow regimes, therefore this limit was considered important.
- Experimental data at flows higher than 3000 kg m^{-2} s^{-1} suggested a different optimized coefficient n than medium flows, hence this limit was chosen as second boundary for mass flux range.
- Different CHF mechanisms are present in subcooled and saturated quality regions, therefore the boundary at $X = 0$ was considered; that evenly divides the subcooled and saturated regions.
- Variation in the optimized coefficient n between different subregions investigated, suggested that the increase of number of subregions would have a positive impact on correlation accuracy but it would make its application more complex; hence the choice of the number of subregions is a compromise.

Within each subregion the exponent n has been computed as a weighted average (the weight of each exponent being proportional with the number of experimental data for each slice) of all available exponents that fall inside the subregion:

$$n = \frac{\sum_{i=1}^{M} N_i n_i}{\sum_{i=1}^{M} N_i} \tag{5}$$

where N_i is the number of experimental data points within the slice i, n_i the computed exponent for the slice i, and M the number of slices considered within the subregion.

The values of optimized exponents n for each range of pressure, quality and mass flux are shown in Table 3.

Table 2
Summary of the diameter correction correlations (Kirillov et al., 1992; Macbeth, 1963).

Prediction Method	Reference	Comments
$\frac{CHF_D}{CHF_{D=8mm}} = \left(\frac{8}{D}\right)^n$	Groeneveld et al (1986)	For 4 <D <16 mm n=1/3 For D > 16mm CHF_D/CHF_{8mm}=0.6
$\frac{CHF_D}{CHF_{D=8mm}} = \left(\frac{8}{D}\right)^{1/2}$	Groeneveld et al (1996)	For 3<D<25 mm P=100-20000 kPa G=0 – 8000 kg m^{-2} s^{-1} Xc= -0.5 to 1
$\frac{CHF_D}{CHF_{D=8mm}} = \left(\frac{8}{D}\right)^{1/2}$	Doroschuck et al (1975)	Valid for 4<D<16 mm for ΔTsub <75 C and not very high qualities
$\frac{CHF_D}{CHF_{D=10mm}} = \left(\frac{10}{D}\right)^n$ $CHF_D=CHF_8(8/10)^{0.6}(10/D)^n$ n=0.4 if D>10mm n=0.6 if D<10mm	Biasi et al (1967)	Ranges : D=3-37.5 mm P=270 -1400 kPa G=100 – 6000 kg m^{-2} s^{-1}
$\frac{CHF_D}{CHF_{D=10mm}} = f(D)$ D: 4, 6, 8, 10, 12, 14, 18, 24, 25 f: 1.18, 1.76, 1.085, 1.00, 0.973, 0.962, 0.94, 0.917, 0.9 Modified Becker correlation for ID=8 mm tubes	Becker (1965)	Ranges: D=4-25mm P=270-10000 kPa G=120 – 5400 kg m^{-2} s^{-1} Xc=0-0.5

Table 2 (*Continued*)

<table>
<tr><td>
<table>
<tr><td>D</td><td>4</td><td>6</td><td>8</td><td>10</td><td>12</td><td>14</td><td>18</td><td>24</td><td>25</td></tr>
<tr><td>f</td><td>1.09</td><td>1.62</td><td>1.00</td><td>0.92</td><td>0.9</td><td>0.89</td><td>0.86</td><td>0.84</td><td>0.83</td></tr>
</table>
</td><td></td><td></td></tr>
<tr><td>$$\frac{CHF_D}{CHF_{D=8mm}} = \left(\frac{8}{D}\right)^{0.1}$$</td><td>Macbeth (1963)</td><td>Valid for low flow (G<300 kg m^{-2} s^{-1})</td></tr>
<tr><td>$$\frac{CHF_D}{CHF_{D=8mm}} = \left(\frac{8}{D}\right)^{0.235}$$</td><td>Mudawar (2000)</td><td>100<p<20000 kPa
300<G<30000 kg m^{-2} s^{-1}
x<-0.05
0.25mm<D<15mm</td></tr>
<tr><td>$$\frac{CHF_D}{CHF_{D=8mm}} = \left(\frac{8}{D}\right)^{n}$$
$$n = 0.58 \cdot (1 - 0.25 \cdot \exp(-2 \cdot x)) \cdot (1 - 15 \cdot D^{-6} G)$$</td><td>Wong (1996)</td><td>For 3<D<25 mm
P=100-20000 kPa
G=0 – 8000 kg m^{-2} s^{-1}
Xc= -0.5 to 1</td></tr>
<tr><td>$$\frac{CHF_D}{CHF_{D=8mm}} = \left(\frac{8}{D}\right)^{n}$$
<table>
<tr><td colspan="4">Exponent n as function of flow region and pressure</td></tr>
<tr><td>P(MPa)</td><td>Region 1</td><td>Region 3</td><td>Region 5</td></tr>
<tr><td>1 - 6</td><td>-0.188</td><td>0.068</td><td>0.266</td></tr>
<tr><td>6 -14</td><td>-0.263</td><td>0.164</td><td>0.197</td></tr>
<tr><td>14 -20</td><td>-0.418</td><td>-</td><td>0.163</td></tr>
</table>
<table>
<tr><td colspan="4">Thermodynamic qualities that define the boundaries of different regions</td></tr>
<tr><td></td><td>X_1</td><td>X_2</td><td>X_3</td></tr>
<tr><td>1<P<6</td><td rowspan="2">$2.7\left(\frac{\rho_g \sigma}{G^2 D}\right)^{0.25}\left(\frac{\rho_g}{\rho_f}\right)^{0.333}$</td><td>$1.05\left(\frac{\rho_g \sigma \cdot 10^3}{G^2 D}\right)^{0.204}\left(\frac{\rho_g}{\rho_f}\right)^{0.214}$</td><td>$1.18\left(\frac{\rho_g \sigma \cdot 10^3}{G^2 D}\right)^{0.238}\left(\frac{\rho_g}{\rho_f}\right)^{0.204}$</td></tr>
<tr><td>6<P<20</td><td>$0.52\left(\frac{\rho_g \sigma \cdot 10^3}{G^2 D}\right)^{0.280}\left(\frac{\rho_g}{\rho_f}\right)^{0.011}$</td><td>$0.57\left(\frac{\rho_g \sigma \cdot 10^3}{G^2 D}\right)^{0.3}\left(\frac{\rho_g}{\rho_f}\right)^{-0.0367}$</td></tr>
</table>
If the dryout quality falls in the transition regions (Regions 2 or 4), then n is calculated through linear interpolation between adjacent regions.

P (MPa), G (kg $m^{-2}s^{-1}$), D (m), ρ (kg m^{-3}), σ(Nm^{-1})</td><td>Boltenko (1989)
Kirillov (1992)</td><td>Region 1: bubbly flow CHF
Region2: transition from bubbly to annular – entrained controlled CHF
Region 3:annular dispersed flow – entrainment controlled dryout
Regon 4: transition form annular entrained controlled CHF to deposition controlled CHF
Region 5: annular dispersed flow – deposition controlled dryout

X_1: upper limit of Region 1
X_2: lower limit of Region 3
X_3:lower limit of Region 5
Region 2 is “between the intersection of CHF curve for bubbly flow with that for annular-dispersed flow and X_1”;
Region 4 between X_2 and X_3</td></tr>
</table>

Table 3
Exponent n as a function of pressure, mass flux and local quality.

Pressure (kPa)	Mass flux (kg m^{-2} s^{-1})	Quality (X)			
		−0.5 to −0.25	−0.25 to 0	0 to 0.5	0.5 to 1
100–14000	0–250	−0.2	−0.2	−0.2	−0.3
	250–3000	0.4	0.4	0.5	0.6
	3000–8000	0.3	0.3	0.4	0.4
14000–21000	0–250	−0.2	−0.2	−0.2	−0.3
	250–3000	0.4	0.2	0.4	0.4
	3000–8000	0.3	0.2	0.2	0.2

Table 4
Average error and RMS errors for different diameter corrections (numbers in italics refer to subcooled data only).

Correlation	# of data	HBM		DSM	
		Avg	RMS	Avg	RMS
This paper (Table 3)	26231/*2055*	-0.03/*-0.5*	7.3/*5.7*	3.7/*-0.5*	38.0/*11.7*
Kirillov(1992)	26231/2055	1.5/*3.8*	9.7/*16.7*	8.6/*8.9*	42.7/*34.9*
Wong(1996)	26231/2055	-0.1/*-0.5*	7.2/*5.9*	3.3/*-0.5*	37.8/*11.9*
Groeneveld(1986) (n=1/3)	26231/2055	0.6/*-0.2*	7.3/*5.8*	5.6/*-0.1*	38.8/*11.8*
Groeneveld(1996) (n=0.5)	26231/2055	-0.2/*-1.1*	7.3/*6.7*	3.4/*-1.3*	37.9/*12.9*
Becker (1967)	26231/2055	2.1/*2.31*	9.1/*9*	11.9/*6.9*	44.5/*20.1*
Biasi(1967)	26231/2055	0.01/*-0.73*	7.4/*6.7*	3.9/*-0.5*	38.3/*13.8*
No correction (n=0)	26231/2055	2.0/*2.03*	9.8/*10.6*	11.3/*4.1*	45.1/*20.4*

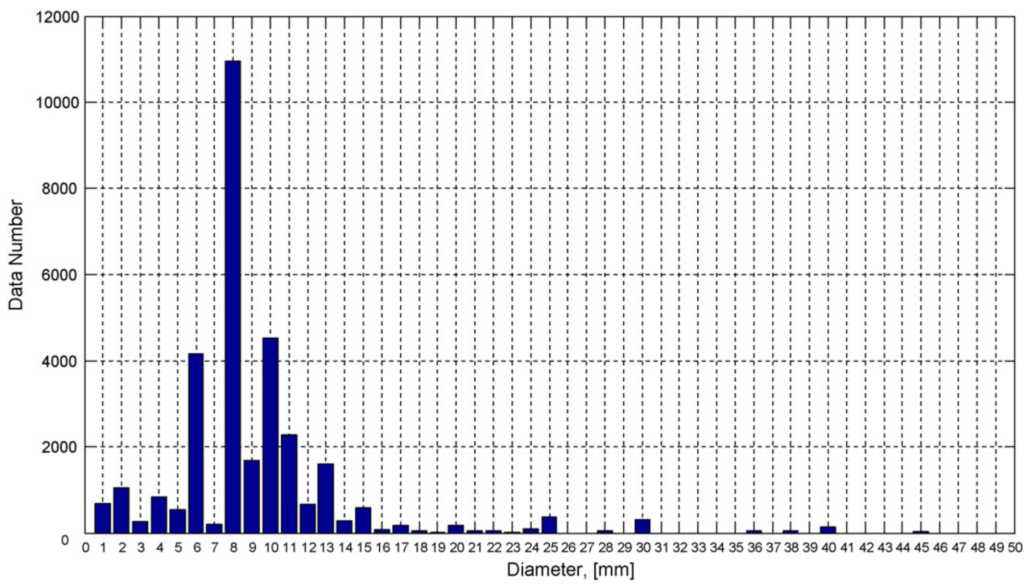

Fig. 2. Distribution of number of AECL-UO experimental data with respect to inside diameter.

5. Assessment of correlations for diameter correction

Seven correlations for diameter correction factor have been assessed against AECL-UO CHF database and the CHF LUT 2005. The results are shown in Table 4.

The case for no diameter correction (last row in Table 4) has been included for comparison purposes. As it can be seen in Table 4, all correlations except Kirillov's and Becker's, provide very similar results, both at constant inlet and local conditions. Wong's correlation provides the lowest RMS errors. Note that constant inlet conditions are sometimes referred as heat balance method (HBM) and constant local conditions as direct substitution method (DSM) (Hejzlar and Todreas, 1996).

Applying no correction for diameter would lead to significantly higher errors than most of the analyzed correction factors. For constant inlet conditions the RMS error is approximately 2.5% higher. For constant local conditions the increase is about 8%. Hence the diameter effect is significant and should be included in any CHF prediction, especially for diameters relatively far from $D = 8$ mm. Note that the experimental data equal or close to 8 mm represent almost 50% of the database, while there are few data with $D > 13$ mm (see Fig. 2).

Kirillov is the only one that assumes that at subcooled dryout conditions a higher diameter results in a higher CHF. This trend is in contradiction with the experimental data and with the other correlations, as well. Table 4 shows that the highest errors of Kirillov's correlation correspond to subcooled and low quality data, where the errors are even greater than the reference case, where no diameter correction was applied.

To further assess the impact of the diameter corrections selected, the CHF LUT was re-derived, using as skeleton table LUT-2005. Different diameter correction equations were used for deriving a new CHF LUT. Hence eight CHF LUTs were derived, each corresponding to one diameter correction method. The error analysis was performed using the same diameter correction method as for derivation. The results are presented in Table 5. It can be observed that the re-derived tables have slightly lower errors than the LUT-2005 skeleton table, for all cases analyzed. Wong correla-

Table 5
Error analysis for the re-derived CHF LUT.

		HBM		DSM	
	# of data	Avg	RMS	Avg	RMS
This paper (Table 3)	26231	-0.43	7.21	0.29	$\boxed{36.43}$
Kirillov(1992)	26231	-0.06	8.64	1.40	38.69
Wong(1996)	26231	-0.43	$\boxed{7.00}$	0.65	37.15
Groeneveld(1986) (n=1/3)	26231	-0.23	7.03	1.12	37.34
Groeneveld(1996) (n=0.5)	26231	0.42	7.06	0.72	37.19
Becker (1967)	26231	0.10	8.40	3.40	40.76
Biasi(1967)	26231	-0.35	7.20	0.92	37.42
No correction (n=0)	26231	0.29	8.57	3.14	40.32

Table 6
Variation of CHF LUT errors with *n*, for HBM and DSM (numbers in italic refer DSM).

	n				
	0.2	0.3	0.4	0.5	0.6
Average	1.2/*7.7*	0.7/*6.0*	0.3/*4.6*	0.2/*3.4*	−0.7/*2.24*
RMS	7.9/*40.5*	7.4/*39.1*	7.1/*38.3*	7.3/*37.9*	7.7/*38.4*

tion provides the lowest RMS error for constant inlet conditions and the correlation proposed by this paper the lowest errors for constant local condition. The variation between most of the methods is small.

6. Discussion

Table 5 shows that Wong (1996), Groeneveld et al. (1986, 1996), Biasi et al. (1967) and the correlation proposed by this paper (Tanase, 2007) provide similar results when compared with AECL-UO CHF experimental database and the CHF LUT 2005. Although Wong's correlation gives the lowest RMS errors, it seems that a constant exponent $n = 0.4–0.5$ satisfactorily predicts the whole range of pressure, mass flux and critical quality of the CHF LUT.

6.1. Effect of variation of exponent n (constant n, for the whole LUT range)

A sensitivity analysis of average and RMS errors with respect to exponent n has been performed. Different exponents n were tested and the results are shown in Table 6.

Minimum RMS error is obtained at $n = 0.4$ for constant inlet conditions, while for constant local conditions the minimum RMS is obtained at values slightly higher than 0.5.

6.2. Final recommendation for diameter correction

Tables 4 and 5 have shown that most diameter corrections give significantly lower errors when compared with the reference case (no correction), hence a diameter correction should be employed when predicting CHF in tubes with IDs far from 8 mm. Very small difference between Groeneveld et al. (1986, 1996), Wong (1996), Biasi et al. (1967) and the correlation proposed by this paper lead to the conclusion that any of them can be successfully applied for most applications. Obviously, a simple one (a constant exponent over the whole LUT range) may be preferred. The analysis from 6.1 has shown that a constant $n = 0.4–0.45$ seems to give the most promising results. However, when a higher degree of accuracy for the diameter correction is required, a more complex correlation (such Wong's) is recommended.

From Table 4 one can notice that, in the subcooled region, a lower exponent (Groeneveld et al., 1986, $n = 0.33$) gives better results. As shown in Table 2, for this region Mudawar (2000) recommends an exponent 0.235. Therefore, in subcooled range, a constant exponent within the range of 0.25–0.33 is recommended.

Another particular aspect is represented by the low flow region ($<250\,\mathrm{kg\,m^{-2}\,s^{-1}}$). As pointed out by Mishima et al. (1985, 1987), in the flooding type CHF (low flow low pressure conditions) the CHF increases with an increase in diameter, therefore a negative exponent $n = -0.2$ to -0.3 is recommended.

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